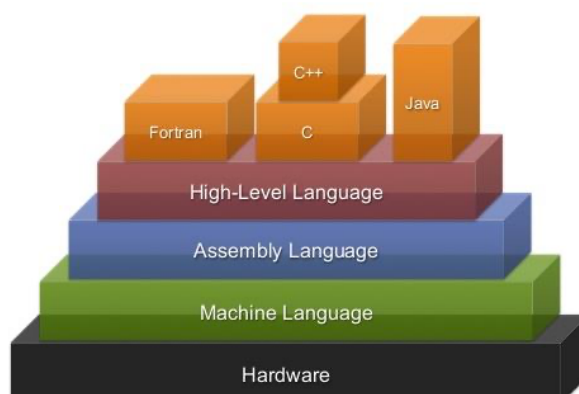
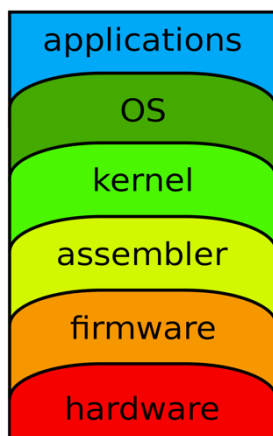


Part 1: Core Computer Architecture

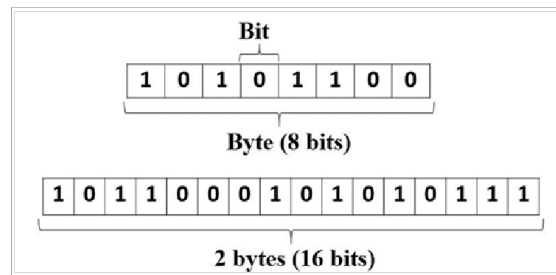
- **Source code:** set of instructions (code) written in human-readable programming language, source code must be compiled into machine code to execute.
- **Machine code:** lowest-level representation of a computer program, in binary, executable by a CPU – will be specific to a processor's architecture.
- **Bytecode:** low-level code (higher than machine code) not specific to hardware. Bytecode is an immediate representation of a program that can be further compiled or interpreted at runtime.
- **Operating System (OS):** Software that manages hardware resources and provides common services for computer programs e.g. memory management, file management/system, process execution. Acts as an intermediary between users and the computer hardware, e.g., Windows, macOS, and Linux.
- **Hardware:** The physical components of a computer system, such as the CPU, memory, storage devices, and peripherals, that perform various computing tasks.
- **Software:** Programs and applications that run on a computer, enabling users to perform specific tasks, such as word processing, web browsing, and gaming.
- **Middleware:** Software that acts as an intermediary layer between operating systems and applications, facilitating communication, data management, and input/output, often used in distributed systems to enable interoperability and data exchange.
- **Firmware:** Low-level software programmed into the read-only memory (ROM) of a hardware device, providing control, monitoring, and data manipulation functions for the hardware. For example the embedded firmware in a laptop's touchpad, or firmware on CPU which controls the process and can be updated to improve performance or fix bugs when OS updates.
- **Kernel:** The core component of an operating system that manages system resources and hardware, facilitating communication between software applications and the hardware.
- **Assembler:** A software tool that translates assembly language, a low-level programming language closely related to machine code, into executable machine code that a computer's CPU can understand and execute.
- **Assembly Language:** A low-level, human-readable representation of machine code that provides a more understandable way to write and debug programs at the hardware level. Assembly language is specific to CPU.



- **Central Processing Unit (CPU):** The main processor of the computer that executes instructions and processing tasks.
- **Storage:** Persistent storage for saving data and applications more permanently e.g. solid-state drives (SSDs) using flash memory – a memory storage medium which retains data in absence of power supply.
- **Random Access Memory (RAM):** Temporary memory used for data storage while a computer is running needed to run applications and tasks. All data here is lost when computer is shut-down.
- **Read-only Memory (ROM):** Persistent memory used in computers and electronic devices to store firmware or software that is rarely changed, retaining data even when the power is turned off e.g. Basic input/output system (**BIOS**) – used to initialise hardware operations on computer boot-up.
- **Graphics Processing Unit (GPU):** Specialised processor for accelerated rendering of images, video and other graphics. Commonly used in modern AI applications to accelerate deep learning tasks such as training neural networks.
- **Compiler:** program that translates programming language (source code) into a machine code executable file that a computer can execute. The resulting machine code can be reused. Typically faster than interpreted programs: translation only once, but translation can be slow. Languages: C, C++, Rust.
- **Compile Time:** The phase where the source code is converted into executable code by a compiler, detecting syntax and type errors.
- **Run Time:** The phase where the compiled code is executed on the machine, performing operations and potentially encountering runtime errors.
- **Interpreter:** program that translates programming language (source code) into machine code line-by-line for execution. Typically slower than compiled programs since line-by-line, no executable file produced. Languages: Python, JavaScript, PHP.
- **Just-in-time compiler (Hybrid approach):** program which first compiles language to intermediate code like bytecode. Bytecode then just-in-time compiled to machine code at runtime rather than prior to execution (by virtual machine e.g. Java Virtual Machine). This allows for both rapid execution start-up similar to interpreted languages and optimized performance akin to compiled languages, adapting optimizations based on actual program execution.
- **Binary Digit (Bit):** most basic unit of information in computing – a “1” or a “0”
- **Binary:** number system with 2 digits 0 and 1, on a base 2 system – meaning each number position (column) represents a power of 2. The binary number 1011 = eleven:

$$(1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0) = 8 + 0 + 2 + 1 = 11$$

- **Byte:** Unit of data which is 8 binary digits long. A byte is the unit most computers use to represent a character such as a letter, number or typographic symbol.



- **Runtime environment:** system where software programs are executed, includes all necessary software (libraries, execution engines) and hardware configurations required to run a program. Provides a layer of abstraction on resource allocation, memory management etc. Examples are NodeJS, Java Runtime Environment (JRE).
- **Execution Engine:** takes compiled or interpreted code and performs the execution on the hardware. Responsible for execution, memory management, garbage collection, just-in-time compilation.
- **JavaScript:** is a dynamic, interpreted programming language commonly used for web development. It is the core technology of the World Wide Web, alongside HTML and CSS, and allows for interactive web pages and applications. JavaScript is characterized by its event-driven, functional programming style.
- **JavaScript engine:** software component that executes JavaScript by parsing it and performing just-in-time compilation for execution on the host system processor. Each browser has its own JavaScript engine e.g. "V8" – Google.
- **Event Loop:** Part of the runtime that schedules tasks and manages the order of execution based on the state of the call stack and the task queues
- **NodeJS:** free and open-source JavaScript runtime environment built using Chrome V8's engine, allowing JavaScript development outside the browser.

Part 2: Programming

- **(Computer) Program:** set of instructions for a computer to execute.
- **Variables:** Variables are used to store data values. They can hold different types of data such as integers, strings, objects, etc.
- **Data Types:** Different programming languages support various data types like integers, floats, strings, booleans, arrays, and objects.
- **Algorithms:** Algorithms are step-by-step procedures or formulas for solving problems. They are fundamental to computer science and programming.
- **Data Structures:** Data structures are ways of organizing and storing data to enable efficient access and modification. Examples include arrays, linked lists, stacks, queues, trees, and graphs.
- **Control Structures:** Concepts like conditionals (if statements), loops (for, while), and switch statements. They control the flow of execution in a program.
- **Functions/Methods:** Code grouped into reusable blocks and can accept parameters (inputs) to carry out some logic.
- **Recursion:** Recursion is a technique where a function calls itself. Understanding recursion is important for solving certain types of problems.
- **Class:** a template definition of the methods and variables for a particular object.
- **Object:** In object-oriented programming an object is an instance of a class (a specific creation of a class), different objects of the same class can have different values for the same properties e.g. class of "Car" with different values for "models" or "mileage". (In languages like JavaScript an object literal is key-value pair enclosed in curly braces {} – different from object as an instance of a class).
- **Object-oriented Programming:** programming paradigm based on the concept of objects. Programs are made by interacting the objects together. OOP paradigm includes concepts like encapsulation, inheritance, polymorphism and more. Benefits of OOP include maintenance, security, and reusability.
- **Clean code:** writing clean code means readable and easily understandable to others e.g. variable naming, avoiding duplication, less code is typically better.
- **Operator:** symbol or function that tells the computer to perform specific mathematical, relational, or logical operations and produce a final result.



- **Arithmetic Operator:** mathematical operations e.g. '+', '-', '*', '/', and '%' (modulus).
- **Logical Operator:** logical operations, primarily in conditional statements, to determine the truthfulness of expressions e.g. 'AND', 'OR', 'NOT'.
- **Bitwise Operator:** operators that perform specific operations on operands (typically integers) to achieve a result.

- **AND (&):** sets each bit to 1 if both bits are 1.
 - **OR (|):** sets each bit to 1 if at least one of the bits is 1.
 - **XOR (^):** sets each bit to 1 if only one of the two bits is 1.
 - **NOT (~):** flips all the bits.
 - **Shift operators (<< and >>):** move bits left or right, filling with zeros or extending the sign bit.
 - **Assignment Operators:** assign values to variables e.g. '=', '+='.
 - **Unary Operators:** Operate on only one operand to change its value '++'
 - **Ternary Operator:** shorthand conditional operator e.g. '?:'
- **Dynamic Typed Languages:** Variables can hold any type of data and the type is determined at runtime. This allows for greater flexibility but can lead to runtime errors if types are misused. Example languages include JavaScript, Python, and Ruby.
 - **Static Typed Languages:** Variables have a fixed type that is determined at compile-time. This allows for type checking before runtime, reducing the chance of type-related errors. Example languages include Java, C++, and TypeScript.
 - **TypeScript:** a superset of JavaScript compiled to JavaScript using a TypeScript compiler. The main goals of TypeScript are to add optional types and implement planned features of future JavaScript.
 - **Npm:** Node package manager, consisting of a command line client and an online database of public and paid private packages – npm registry. Npm allows developers to easily install/update packages defined in the package.json file. Npm is included in the Node.js runtime.
 - **Package.json:** any Node.js runtime will include a package.json file which is the manifest for the project, including the packages/library dependencies, their version, project metadata.
 - **Semantic versioning:** Software version format of the style X.Y.Z, being major, minor, patch releases.
 - **Exception Handling:** Exception handling is the process of responding to and recovering from errors in a program. This includes try-catch blocks to handle exceptions gracefully.
 - **Debugging:** Debugging is the process of finding and fixing errors or bugs in a program. Knowing how to use debugging tools and techniques is essential for any programmer.

Part 3: Software development

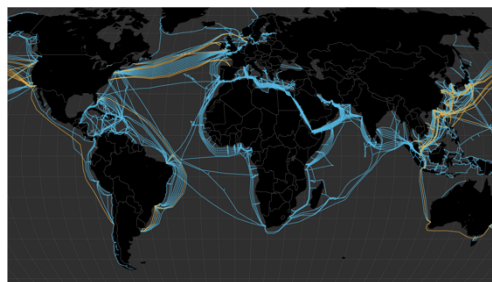
- **Software Development Life Cycle (SDLC):** Understanding the phases of software development, including requirements gathering, design, implementation, testing, deployment, and maintenance.
- **Version Control:** Version control systems like Git allow developers to track changes to their codebase, collaborate with others, and revert to previous versions if needed.
- **Git:** open-source version control system for tracking changes in a set of files. Git is distributed meaning that every developer has a local copy of the entire project history, this allows developers to work locally. Using Git we can:
 - **Clone:** Copying a repository to create a local version on your computer.
 - **Commit:** Saving a snapshot of your changes to the local repository.
 - **Push:** Sending local commits to a remote repository.
 - **Pull:** Fetching changes from a remote repository and integrating them into the local environment.
 - **Branch:** Creating a separate line of development.
 - **Merge:** Combining changes from different branches back into a single branch.
- **Github:** web-based platform that facilitates version control using Git. Github does this by providing a cloud platform for hosting Git repositories.
- Debugging and Testing:
- **Technical debt:** The implied cost of additional rework caused by choosing an easy solution now instead of a better approach that would take longer or natural build-up of complexity in the code.
- **Waterfall:** Waterfall is a linear and sequential software development methodology where each phase (requirements, design, implementation, testing, deployment, maintenance) must be completed before the next phase begins.
- **Agile:** A methodology focused on iterative development, where requirements and solutions evolve through collaboration between cross-functional teams.
- **Scrum:** An Agile framework that uses fixed-length iterations called sprints, with defined roles (Scrum Master, Product Owner, Development Team) and ceremonies (daily stand-ups, sprint planning, reviews, and retrospectives).
- **Kanban:** A visual workflow management method used to optimize and manage the flow of work, emphasizing continuous delivery and efficiency by visualizing tasks on a Kanban board.
- **Software Design Patterns:** Design patterns are reusable solutions to common problems in software design. They help improve code readability, scalability, and maintainability, e.g. Strategy, Singleton, Factory Method.
- **Unit Testing:** Automated testing of individual units or components of a software, mocking external dependencies (libraries) to work as expected e.g. testing a function that calculates the total price of items in a shopping cart to ensure it returns the correct sum.
- **Integration Testing:** Automated testing of the integration of different modules or components e.g. a user authentication module and the database to ensure that user credentials are correctly verified.

- **System/End-to-End Testing:** Automated testing of the complete integrated system to verify that it meets the specified requirements e.g. entire user registration process from filling out the sign-up form to receiving a confirmation email to ensure the workflow functions as expected.
- **Code Review:** The practice of reviewing code by other developers to identify mistakes overlooked in the initial development phase, improve code quality, and ensure consistency.
- **Refactoring:** The process of restructuring existing computer code without changing its external behavior to improve readability, reduce complexity, and improve maintainability.
- **Test-Driven Development (TDD):** A software development process where tests are written before the code that needs to pass the tests.
- **DevOps:** set of practices that combines software development (Dev) and IT operations (Ops) – aimed at the continual improvement of the efficiency of the SDLC. Efforts are focused on continuous integration and continuous delivery, for example automating pipelines for testing and deploying code changes.
- **Continuous Integration and Continuous Delivery (CI/CD):** software development practice where code changes are automatically tested and merged (continuous integration), and then automatically deployed to production environments (continuous deployment).
- **Database:** organised collection of data that is stored and accessed electronically, designed to manage, store, and retrieve large amounts of information efficiently.
- **Relational Database:** type of database that uses a structured format of tables to store data with pre-defined relationships, using SQL (Structured Query Language) for querying and managing the data. Examples include MySQL, PostgreSQL, and Oracle.
- **Non-relational Database:** type of database designed to handle unstructured or semi-structured data, offering flexible schema designs and scalability at the expense of structure. Examples include MongoDB, Cassandra, and Redis, data is typically stored as JSON.
- **Object-Relational Mapping (ORM):** software/programming technique that translates data representations used by databases to programming objects used in OOP.
- **Structured Query Language (SQL):** programming language used to manage and manipulate relational databases by performing tasks such as querying, updating, and managing data.
- **Database Management System (DBMS):** software that provides an interface for users to create and interact with databases, manage data, and enforce data integrity. For example managing the reading and writing of physical data storage in memory, access and authentication, backup and recovery etc. Examples include MongoDB, Amazon RDS, Oracle, MySQL.
- **PostgreSQL:** An advanced, open-source relational database management system (RDBMS) known for its robustness and support for complex queries, transactions, and various data types.
- **Containers:** Containers are lightweight, portable units of software that package an application and its dependencies, ensuring consistency across different computing environments by abstracting the differences in underlying host OS via the container runtime environment.

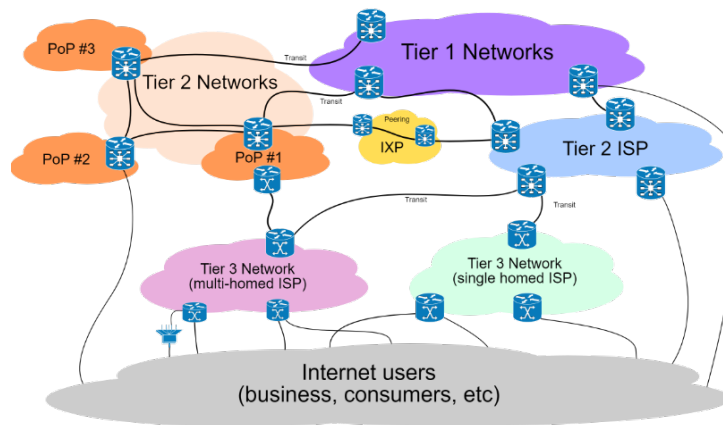
- **Docker:** Docker is an open-source platform and tool that uses containerization technology to automate the deployment, scaling and management of containerised applications. Inside a Docker container, Docker acts as a runtime environment, for example NodeJS runs JavaScript apps, JRE runs Java apps, Docker runs containerised applications.
- **Docker Image:** lightweight, standalone, read-only, executable template that contain instructions for creating a container. A snapshot of the application and dependencies needed inside a container for the software to run.
- **Application Programming Interface (API):** software/service/protocol that allows different software applications to communicate and interact with each other using requests and responses. For example, on your browser you send a request to Facebook when creating an account, with a body of data {name: "Smith", age: "30"}, to an API endpoint on Facebook's servers.
This endpoint can look like www.facebook.com/api/user/create for example. It receives the requests, creates a user in database internally and returns a successful response to let the us (the client) know it worked ok.
- **State:** the stored information or condition of a system or application at a specific point in time.
- **Service-Level Agreement (SLA):** A contract between a service provider and a customer that specifies the expected level of service, performance, and availability.

Part 4: Networking and the Internet

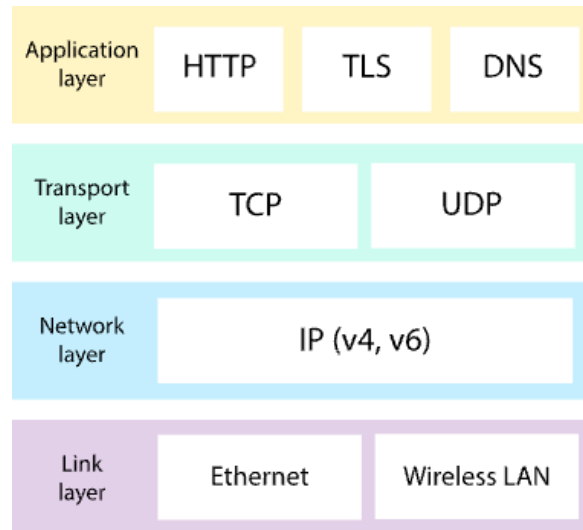
- **Network:** A collection of interconnected devices that can communicate with each other to share resources and information. They communicate using standardised protocols and some transmission medium e.g. phone using Wi-Fi – radio waves (transmission medium) as well as the Wi-Fi (IEEE 802.11) protocol.
- **Local Area Network (LAN):** A network that connects devices within a single, limited area, such as a home, school, or office building. Can be wired or wireless, usually privately owned and managed e.g. home network connecting computers, phones and printers, or office building.
- **Wide Area Network (WAN):** A network that connects devices over a large geographical area, often using leased telecommunication lines or satellite links. WANs connect multiple LANs, and are usually managed by service providers or organisations e.g. the internet, a bank's network connecting ATM's across country, a company's network connecting offices in different cities.
- **Server Client Model:** fundamental concept in computer networking that describes how different devices communicate and interact within a networked environment, such as the internet.
- **Server:** computer (can refer to physical hardware or the software) that provides services/resources to other computers, known as clients, in the network. Important examples are web servers (hosting websites), or database servers (hosting data and access to it through queries).
- **Client:** device or software application that requests and utilizes services, resources, or data from a server, e.g. web browsers on your laptop (to access websites hosted on servers).
- **Internet:** Global network that communicates using the internet protocol.
 - **Internet Backbone:** Consists of global **fiber optic cables** (thin, flexible glass core fibre wires through which light can be sent) and infrastructure, often referred to as "global pipelines," that interconnect continents and major cities.



- **Tier 1 Network:** Internet Protocol (IP) network that can reach every other network on the Internet without paying for access/purchasing IP transit. Physically we can consider this the backbone of the internet, the main routes for data traffic between continents: fibre-optic cables across oceans and land, data centers, internet exchange points. Tier 1 networks are set up by large internet service providers e.g. AT&T and BT.

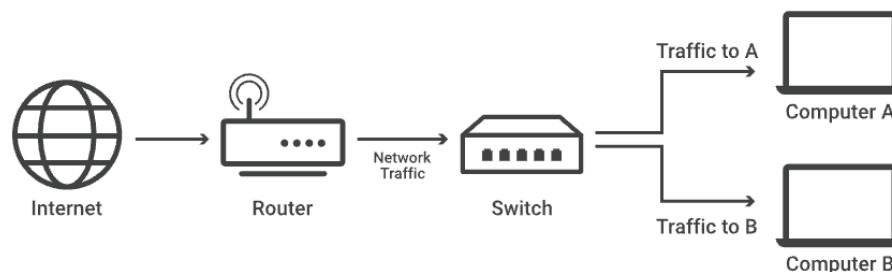


- **Tier 2 Network:** ISP that connects to internet through peering and purchasing transit from Tier 1 networks, and providing paid access to internet e.g. Comcast, Vodafone. Tier 2 typically have a regional to global reach, providing services to enterprises, Tier 3's, and sometimes residential.
 - **Peering:** the process of two or more internet networks connecting and exchanging traffic – physically will occur at internet exchange points. If peering between networks was not possible the internet would segmented per ISP – operating as an isolated network. Peering is “settlement-free” – meaning neither ISP pays the other, instead keeping revenue from own customers.
 - **Internet Exchange Point (IXP):** physical infrastructure through which ISPs exchange internet traffic between networks.
 - **Tier 3 Network:** Network that mainly purchases transit from other networks to the internet to provide local connectivity to end-users e.g. Plusnet (UK) or Spectrum (US). Differs from tier 2 in that tier 3 primarily purchase from ISPs and focus on selling to end-users (residential and small businesses).
 - **(ISP) Point of Presence (POP):** physical location (data centre) where network equipment (routers, switches etc.) is housed to provide local access to a network e.g. Comcast data centre in New York to allow local customers to reach Comcast's Tier 2 network.
- **Protocol:** set of rules and standards that define how data is transmitted and received over a network e.g. Wi-Fi protocol specifies how wireless devices communicate within a LAN.
 - **Packet:** generic term for a formatted unit of data transmitted over a packet-switched network, containing the data to be sent (payload) and the metadata (header) needed for routing and reassembly at destination.
 - **Datagram:** specific type of packet. Self-contained, independent packet of data with sufficient information to be routed from source to destination. A basic unit of transfer
 - **Internet Protocol Suite (IP Suite):** set of communication protocols (the framework) used in the Internet, including protocols – IP TCP, UDP, and others for data transmission, addressing, routing, error-handling etc across networks.



- **Internet Protocol (IP):** network communication protocol in the IP suite for relaying datagrams across networks boundaries. IP covers the delivery of packets from host to destination based on IP addresses. Packets have IP address and other information attached to help routers send them to the correct destination (no information on where to direct inside destination).
- **Transmission Control Protocol (TCP):** protocol in the IP suite – operating over IP, ensuring reliable, ordered transport of data between applications. TCP uses segments as its packets format, a TCP segment (header and payload) is encapsulated within the payload of an IP datagram. Packets are often too small to contain the entire data e.g. a webpage, so TCP ensures that various packets get reassembled correctly without packet loss. TCP requires a connection between sender and receiver, establishing the connection through a “three-way handshake” procedure.
- **Hypertext Transfer Protocol (Secure) HTTP(S):** communication protocol in the IP suite defining the transfer of hypertext documents across the World Wide Web, the format, and expected behaviour of server or client. HTTPS is secure version of HTTP where data is encrypted using Transport Layer Security (TLS).
- **IP Addresses (Internet Protocol Addresses):** Unique numerical labels assigned to each device connected to a network, enabling identification and communication. For example, your home router is assigned an IP address by the ISP used to communicate to external networks. The router also has a private IP address for its local network used to manage the local network. The router uses DHCP to assign private IP address to devices in the local network.
- **Dynamic Host Configuration Protocol (DHCP):** network management protocol used to dynamically assign an IP address to any device or node in a network to enable communication using the internet protocol e.g. DHCP is widely used in LAN - in a home setup the router acts as DHCP server, when a device connects to Wi-Fi, the device sends a DHCP request to router.
- **Router:** Devices that forward data packets between computer networks, connecting networks e.g. LAN and WAN, using IP addresses to find destination.
- **Network Interface Controller (NIC):** hardware component that allows a computer or device to connect to a network e.g. built-in radio to communicate over Wi-Fi.

- **MAC Address (Media Access Control Address):** unique identifier assigned to a network interface card (NIC) for communications at the data link layer (see OSI).
- **Switch:** network device that connects multiple devices within a single LAN using MAC addresses to forward data packets to the correct destination. Modern consumer routers will have built-in switches.

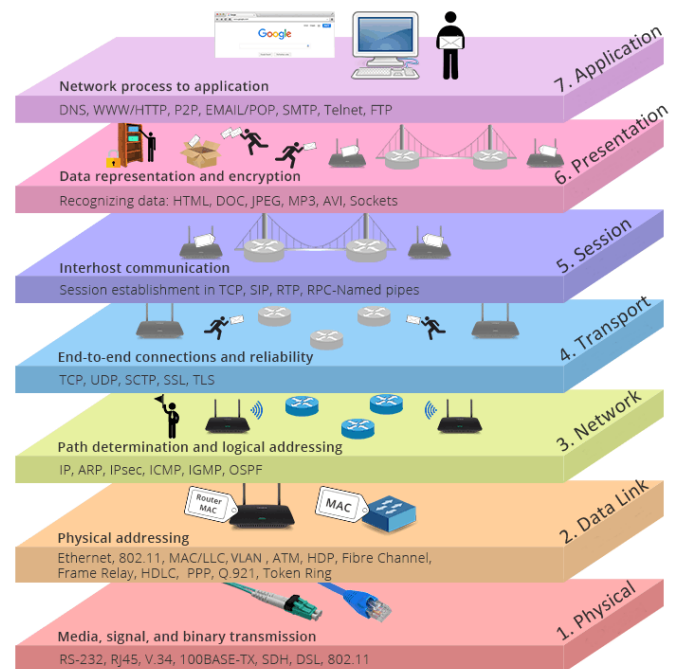


- **Different types of internet connectivity options:**

Feature	Fiber-Optic	Broadband (DSL/Cable)	Cellular (4G/5G)	Satellite
Speed	Very high (up to 1 Gbps or more)	High (up to 100 Mbps)	Moderate to high (up to 1 Gbps with 5G)	Moderate (up to 100 Mbps)
Latency	Very low	Low to moderate	Moderate	High
Reliability	Very reliable	Reliable	Variable	Variable
Coverage	Urban and suburban areas	Widely available	Wide (urban, suburban, rural)	Almost everywhere, including remote areas
Installation	Requires fiber lines	Uses existing phone/cable lines	No installation needed for mobile devices	Requires satellite dish
Cost	High initial cost, moderate ongoing	Moderate to high	Varies with data plans	High (equipment and service)
Use Cases	High-demand applications (streaming, gaming)	General home and business use	Mobile internet, remote access	Remote areas, backup connectivity

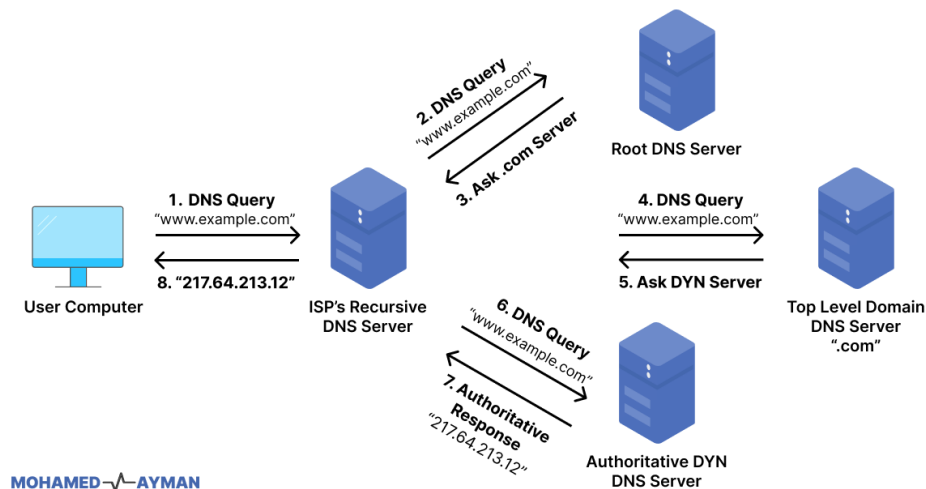
- **Open Systems Interconnection Model (OSI Model):** conceptual framework that standardises the layers of telecommunication into seven functions.

7	Application Layer	Human-computer interaction layer, where applications can access the network services
6	Presentation Layer	Ensures that data is in a usable format and is where data encryption occurs
5	Session Layer	Maintains connections and is responsible for controlling ports and sessions
4	Transport Layer	Transmits data using transmission protocols including TCP and UDP
3	Network Layer	Decides which physical path the data will take
2	Data Link Layer	Defines the format of data on the network
1	Physical Layer	Transmits raw bit stream over the physical medium



- **Domain Name System (DNS):** a name service - service that maps the names of network resources to their network addresses. DNS manages the naming system and mapping database for devices/resources on the Internet. DNS translates human-readable domain names e.g. www.example.com into IP address (192.0.2.1), acting as the internet's directory service or phonebook.
- **DNS Server:** server that hosts the DNS service, storing records of domain names and their IP addresses. DNS servers play a crucial role in directing internet traffic by resolving domain names.
 - **DNS (Recursive) resolver/client:** component in a device that receives DNS queries from client devices, query other DNS servers to resolve the domain, and can cache the results. In your home router the DNS resolver is typically built-into the firmware.
 - **Root DNS Server:** highest level in DNS heirarchy. There are 13 root servers worldwide managed by internet networking organisations. These servers contain global list of top-level domain (TLD) servers and direct queries to these servers.
 - **TLD Server:** servers that store information on the domains with specific top-level domain e.g. .com, .org, .net, managed by large internet companies and entities e.g. Verisign (.com and .net) and the Public Internet Registry (.org). TLD servers provide the authoritative name server IP address.
 - **Authoritative DNS Servers:** servers that provide answers to DNS queries from their own data without needing to query another server. Owned by the organisations that is managing the specific domain. For example if you

register a website with Namecheap and use Namecheap's hosting service – Namecheap will operate the authoritative DNS server for your website, others include AWS, Cloudflare.



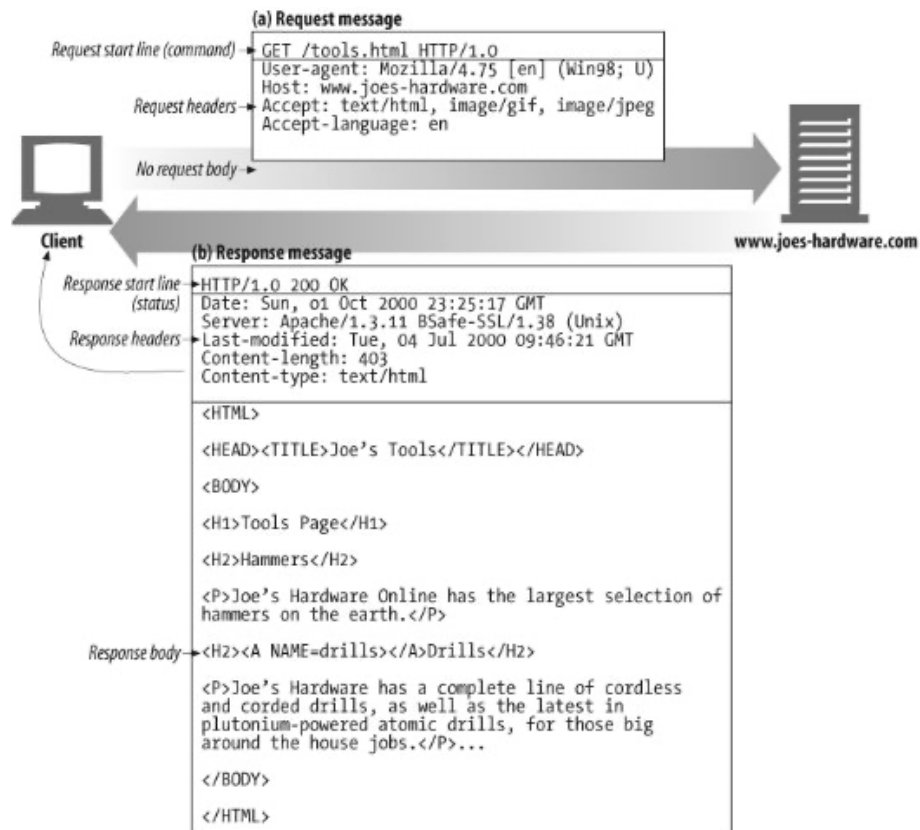
DYN refers to a DNS hosting provider Dynamic Network Services Inc.

- **Firewall:** network security system (software-based) that monitors incoming/outgoing traffic and blocks specific traffic based on security rules. For example on a personal computer it can be a third-party software (McAfee), or on the OS-level (macOS Application Firewall).
- **Ports:** software-based virtual point where network connections start and end. Each port is associated with a specific process or service – allowing computers to differentiate between types of traffic. For example, emails use ports like 993 – Internet Message Access Protocol (IMAP over SSL) while web traffic uses port 80 – HTTP or 443 – HTTP over SSL (HTTPS). All these ports operate over the internet, using the TCP/IP protocol suite.
- **Bandwidth:** measures maximum data transfer rate of a network e.g. bits per second (bps) or megabits per second (mbps), megabytes per second (MB/s), 1MB/s = 8Mbps.
- **Latency:** measures the delay in data transmission, milliseconds (ms).
- **Ping :** network software utility program that tests the reachability of a host on the Internet Protocol. Available most computer operating systems, ping measures the round-trip time for messages sent from the host to a destination computer and back. Pinging involves sending an Internet Control Message Protocol (ICMP) packet, and waiting for the ICMP echo reply.
 - > "ping www.google.com"
 - > 64 bytes from 216.239.38.120: icmp_seq=0 ttl=55 time=16.877 ms

- **Traceroute:** a network diagnostic tool used to track the path and measure the transit delays of packets across an IP network, identifying each hop along the route from the source to the destination.
- **Virtual Private Network (VPN):** service that provides a secure, encrypted connection between a device and a network over an insecure communication medium such as public internet. A VPN can mask IP address by routing data through a remote server that can also bypass geographic restrictions.
- **Packet Switching:** method used for data transmission in which data is broken into small packets that are transmitted independently over various, optimal network paths and reassembled at the destination.
- **Secure Sockets Layer (SSL):** deprecated cryptographic protocol for establishing encrypted links over computer networks commonly for securing web traffic, largely replaced by TLS.
- **Transport Layer Security (TLS):** cryptographic protocol evolved from SSL providing secure communication between client and server in a network.
- **Secure Shell (SSH):** cryptographic network protocol for securely operating network services over an unsecured network e.g. remote login and command-line execution on another computer.

Part 5: Web Development

- **Web Development:** The process of creating and maintaining websites and web applications, encompassing design, software development, hosting etc.
- **World Wide Web (Web):** the information system of content (hypertext documents and other multimedia resources) accessed via the internet (through Hypertext Transfer Protocol - HTTP). Resources are located using URLs.
- **Web Browser:** Software application used to access websites on the World Wide Web, such as Chrome and Firefox. The browser is responsible for retrieving files from a web server and displaying them on user's screen using a rendering engine.
- **Web Server:** server that accepts HTTP(S) requests, and responds with respective content that it hosts. The requested content can be existing like a static file, or dynamic, formed on request using software.
- **Hypertext:** Text displayed on a computer or other electronic device with references (hyperlinks) to other text that the reader can immediately access, typically activated by a mouse click, key press, or screen touch.
- **Hypertext Documents:** Digital documents that contain links (hyperlinks) to other documents or resources, allowing users to easily navigate between related information on the web.
- **HTTP (HyperText Transfer Protocol):** protocol used for transmitting hypertext requests and information on the internet, enabling communication between web browsers and servers. Application layer of the IP suite.
- **HTTP Messages:** format for communication between a client and server on the web, consisting of requests made by the client and responses sent by the server, including headers and body content.



- **HTTP Method:** A standardized set of request methods used in HTTP (Hypertext Transfer Protocol) to specify the desired action to be performed on a resource, such as GET, POST, PUT, DELETE, and PATCH.

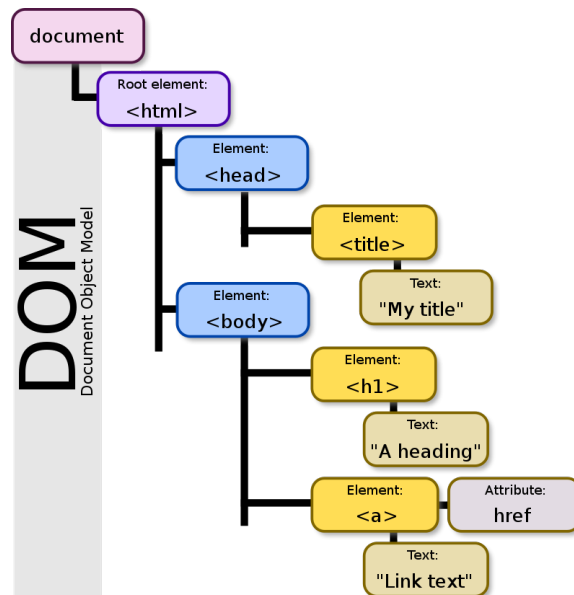
Method	Description
GET	Request for resource from server
POST	Submit data to the server
HEAD	Same as GET but does not return the body
PUT	The data within the request must be stored at the URL supplied, replacing any existing data.
DELETE	Delete a resource
OPTIONS	Return the HTTP methods supported by the server
CONNECT	Client requests the HTTP proxy to forward a TCP connection to some destination. Used to create a TCP/IP tunnel for secure connections using HTTP proxies.

- **HTTP Status Code:** Numeric codes sent by a web server in response to a client's request, indicating the status of the request (e.g., success, error, redirection), such as 200 (OK), 404 (Not Found), and 500 (Internal Server Error).

Table 3-2. Status code classes		
Overall range	Defined range	Category
100-199	100-101	Informational
200-299	200-206	Successful
300-399	300-305	Redirection
400-499	400-415	Client error
500-599	500-505	Server error

- **URL (Uniform Resource Locator):** address used to reference and access resources on the Web, specifying the location of a resource and the protocol used to retrieve it, typically structured as **Scheme/Address/Resources** = <https://google.com/maps>. Domain Name Service (DNS) translates the URL into an IP address for locating the web server hosting the resources. URL is a type of URI.
- **URI (Uniform Resource Identifier):** unique string of characters used to identify a resource (abstract – not necessarily in a network can be physical).
- **Hypertext Markup Language (HTML):** the standard language used to create documents to be displayed on a web browser. HTML defines the content and structure of the web content, e.g. `<title>This is a title</title><img src=https://example.com/image1 width="50" height="50">`
- **Cascading Style Sheets (CSS):** a style sheet language used to describe the styling of a document written in HTML e.g. `.title { background-color: black }`
- **JavaScript:** is a dynamic, interpreted programming language commonly used for web development. It is the core technology of the World Wide Web, alongside HTML and CSS, and allows for interactive web pages and applications. JavaScript is characterized by its event-driven, functional programming style.
- **Frontend:** the section of web development concerned with the development of the graphical user interface (GUI) using technologies like HTML, CSS, and JavaScript.
- **Responsive Design:** The approach to web design that makes web pages render well on a variety of devices and window or screen sizes, ensuring a good user experience across different devices.

- **Document Object Model (DOM):** the virtual model that represents the structure of a document, such as the HTML representation of a web page.



- **Web APIs:** Interfaces that allow different software applications to communicate and interact with each other over the web.
- **JavaScript Object Notation (JSON):** a lightweight string format for storing and transporting data, human-readable, and, commonly used for transmitting data in web applications. Example:

```
{
  "employees": [
    { "firstName": "John", "lastName": "Doe" },
    { "firstName": "Anna", "lastName": "Smith" },
  ]
}
```

- **(Web) Frameworks:** Predefined structures and tools for developing web applications, streamlining development with reusable code and components, examples include NextJS, and Django.
- **React:** a popular, open-source JavaScript library developed by Facebook for building user interfaces, that enables developers to create reusable UI components.
- **Backend:** the server-side part of a software application responsible for processing business logic, database interactions, authentication, and API communication, typically handling tasks that are not visible to the user.
- **Fullstack:** the software development approach combining both frontend (client-side) and backend (server-side) technologies.
- **Cybersecurity:** The practice of protecting systems, networks, and data from digital attacks, unauthorized access, damage, or theft through various technologies, processes, and controls.

- **Authentication and Authorization:** processes for verifying user identity (authentication) and granting permissions based on that identity (authorization), often implemented using technologies like OAuth, JWT.
- **Open Authorization (OAuth):** a system that allows third-party services to carry out authentication, exchanging user information without exposing user credentials, enabling secure delegated access by granting tokens to clients. For example using Google to sign-up to eBay or another website.
- **JSON Web Token (JWT):** JSON object used for securely transmitting information between parties. A JWT consists of three parts: header, payload (which can include sensitive user info) and a signature. The signature is created by combining the header and payload, and hashing with a secret key (typically on server, so server can verify client). JWTs are stored as HTTP cookies on client-side, providing stateless authentication, since the server does not have to maintain a session but only verify on request. HTTP requests will contain the cookie in the header – allowing the server to verify the JWT with the server secret key. For example when verified this allows server access to user info (email/pass) and authorise access to protected pages/resources.
- **Cookie:** A small piece of data stored on the user's browser by a website, used to remember information about the user such as login status, preferences, and tracking information, facilitating a personalized and consistent web experience across sessions.
- **Content Management Systems (CMS):** a software application that enables users to create, edit, manage, and publish digital content, typically for websites, without needing extensive technical knowledge, e.g. WordPress.
- **Web App:** A software application that runs on a web server and is accessed through a web browser, enabling users to perform tasks or interact with content over the internet.
- **Web Performance Optimization (WPO):** techniques and best practices for improving the loading speed and performance of web pages, such as minimizing HTTP requests, optimizing images, and leveraging browser caching.
- **CRUD:** An acronym for Create, Read, Update, Delete, representing the four basic operations that can be performed on data in a database or data storage system. CRUD is also used to describe user interface conventions e.g. interacting with social media, users can primarily create posts, read (view) other posts, update and delete.
- **Software as a Service (SaaS):** a method of providing cloud-based software applications over the internet, allowing users to access and use through a web browser without needing to install or maintain the software on their local devices (typically monthly subscription over lifetime payments). Examples include Salesforce, Microsoft Excel.
- **Representation State Transfer (REST):** software architectural style for networked (web) applications, where the core principle is stateless communication (using standard HTTP methods), for example a request to API does not hold information (or state) that will affect its behaviour after the communication is complete.
- **WebSockets:** A protocol that enables two-way, real-time communication between a client and a server over a single, long-lived connection, often used in applications like chat apps and live updates.

Part 7 Cloud Computing

- **Cloud (Cloud computing):** a network of remote servers hosted on the internet used to provide computer resources such as computing and data storage, providing scalable resources and services on demand, without direct active management by the user.
- **Cloud Providers:** Companies that offer cloud computing services, such as Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), and IBM Cloud.
- **Infrastructure as a Service (IaaS):** a cloud computing service model that provides computing infrastructure (hardware) over the internet, such as compute and storage. Customers are responsible for maintaining the infrastructure e.g. OS, middleware etc.
- **Platform as a Service (PaaS):** a cloud-computing service that provides a platform (hardware and software) allowing developers to more easily build, deploy, and manage applications through the cloud. PaaS typically provides a pre-configured framework for a specific task e.g. AWS Elastic Beanstalk – upload code and Beanstalk automatically handles deployment, load balancing, auto-scaling etc.
- **Public Cloud:** cloud computing services offered by third-party providers over the public internet, available to anyone who wants to purchase or lease them e.g. Azure or AWS.
- **Private Cloud:** cloud computing services used exclusively by a single organization, providing greater control over resources and security, often hosted on-premises or by a third-party provider.
- **Virtual Private Network (VPN):** service that provides a secure, encrypted connection between a device and a network over an insecure communication medium such as public internet. A VPN can mask IP address by routing data through a remote server that can also bypass geographic restrictions.
- **Cloud Storage:** a cloud service model in which data can be stored, managed, and backed up remotely e.g. AWS S3.
- **Virtualisation:** The process of creating multiple simulated environments or resources from a single physical hardware system by isolating the shared resources.
- **Virtual Machines (VMs):** emulations of physical computers that run an operating system and applications, providing the functionality of a physical computer. In cloud services VMs are considered a compute resource. A physical host machine or server can run multiple VMs simultaneously using virtualisation technology.
- **Hypervisor:** software that enables virtualisation by allowing multiple virtual machines (VMs) to run on a single physical hardware system, managing and allocating resources to each VM.
- **Serverless Computing:** a cloud computing execution model where the cloud provider dynamically manages the allocation of machine resources, charging based on actual usage rather than pre-purchased capacity. For example using AWS API Gateway to create a RESTful API that triggers AWS Lambda (serverless functions that automatically run code on request).
- **Elasticity:** The ability of a cloud service to automatically scale resources up or down based on demand, ensuring optimal resource usage and cost-efficiency.